A Scalable Hybrid Quantum-Classical Framework for Oil Reservoir Well Placement Optimization via QUBO Modeling and QAOA

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Abstract

Optimal well placement in oil reservoirs is a complex combinatorial optimization problem that directly affects production efficiency and economic return. Classical approaches can provide strong solutions for small problem sizes, but their scalability becomes increasingly limited as the search space grows. This study presents a scalable hybrid quantum-classical framework for oil reservoir well placement optimization based on Quadratic Unconstrained Binary Optimization (QUBO) modeling and the Quantum Approximate Optimization Algorithm (QAOA).

The proposed framework formulates the reservoir optimization problem as a binary quadratic model that balances production maximization and interference minimization. Its performance is benchmarked against exact optimization, simulated annealing, and random search across multiple candidate-well configurations. Experimental results show that QAOA consistently achieves near-optimal objective values, competitive production levels, and effective interference control. Although classical exact solvers remain faster for small-scale instances, the proposed approach offers a promising scalable pathway for larger reservoir optimization problems where exhaustive methods become computationally infeasible.

Keywords: Oil Reservoir Optimization, Well Placement, QUBO, QAOA, Quantum Computing, Hybrid Optimization

1 Introduction

Optimal well placement is a central problem in reservoir engineering because it strongly influences field productivity, interference behavior, and long-term recovery. The problem is combinatorial in nature, since each candidate location may either be selected or rejected, and the number of feasible configurations grows exponentially with the number of wells. As a result, well placement optimization rapidly becomes computationally challenging.

Classical methods such as brute-force enumeration, simulated annealing, and random search have long been used to address similar decision problems. Exact methods guarantee optimality, but they do not scale well. Heuristic techniques offer practical alternatives, though they may not always preserve near-optimal performance under increasing complexity.

Recent progress in quantum computing has introduced new tools for combinatorial optimization. Among them, the Quantum Approximate Optimization Algorithm (QAOA) has attracted substantial interest due to its compatibility with binary optimization problems expressed in Quadratic Unconstrained Binary Optimization (QUBO) form.

In this study, a hybrid quantum-classical framework is proposed for oil reservoir well placement optimization. The framework combines QUBO modeling, classical baselines, and QAOA-based optimization to investigate the potential of quantum-inspired reservoir decision support.

2 Problem Formulation

Consider a reservoir with N candidate well locations. Let the binary decision variable be defined as:

$$x_i \in \{0, 1\}, \quad i = 1, 2, \ldots, N \tag{1}$$

where $x_i = 1$ indicates that candidate well i is selected, and $x_i = 0$ otherwise.

The objective is to maximize production while minimizing interference between selected wells. This can be written as:

$$\max \left(\sum_{i=1}^{N} P_i x_i - \lambda \sum_{i<j} I_{ij} x_i x_j \right) \tag{2}$$

where:

- P_i is the production potential associated with well i,
- I_{ij} is the interference penalty between wells i and j,
- λ is a weighting factor controlling the trade-off between production and interference.

This objective can be reformulated as a QUBO problem:

$$\min x^T Q x \tag{3}$$

where Q is the QUBO matrix that encodes both production rewards and interference penalties.

3 Methodology

3.1 Classical Optimization Methods

Three classical baselines are considered in this study:

- **Exact optimization**: exhaustive enumeration of all feasible binary well configurations.
- **Simulated Annealing (SA)**: a probabilistic heuristic approach for near-optimal search.

- **Random search**: a baseline method used for comparison.

3.2 Quantum Optimization via QAOA

The Quantum Approximate Optimization Algorithm (QAOA) is employed to solve the QUBO problem using a hybrid quantum-classical optimization loop. The QUBO objective is mapped into a Hamiltonian representation, and the circuit parameters are iteratively optimized to approximate low-energy solutions.

3.3 Data Generation

Synthetic reservoir instances are generated using simplified production and interference patterns. These instances allow consistent benchmarking of all solvers under controlled conditions while preserving the combinatorial structure of the well placement problem.

4 Experimental Setup

Experiments were conducted for three problem sizes corresponding to:

$$N = 6,\ 8,\ 10 \tag{4}$$

For each problem size, multiple runs were performed using different random seeds. The following evaluation metrics were recorded:

- Objective energy
- Production value
- Interference penalty
- Runtime

5 Results and Discussion

5.1 Objective Energy Comparison

Figure 1 compares the objective energy values across the tested methods. Lower values indicate better optimization performance. The results show that QAOA achieves objective values close to those obtained by exact optimization and simulated annealing, indicating near-optimal behavior.

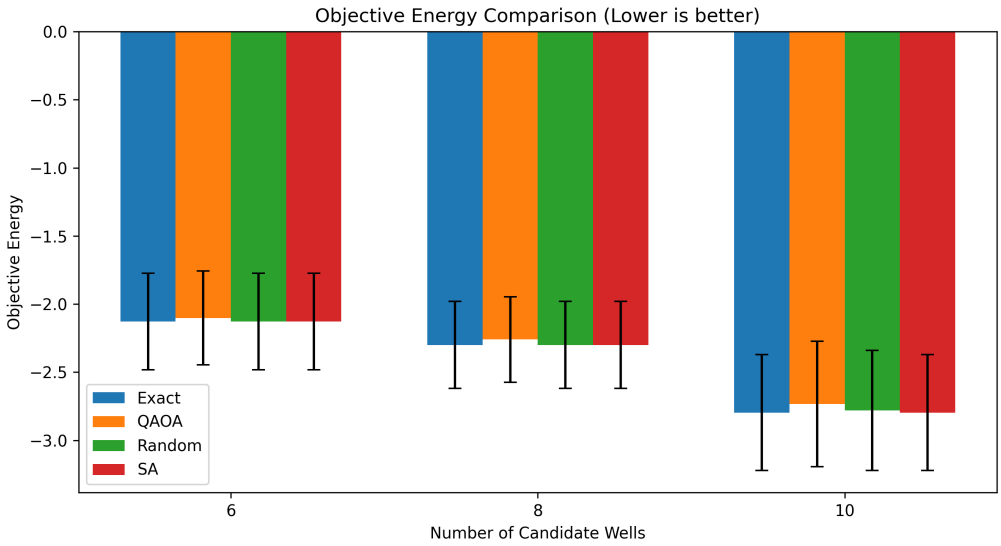

Figure 1: Objective energy comparison across optimization methods (lower is better).

5.2 Production Comparison

Figure 2 presents the production performance. Higher values are preferred. QAOA maintains production levels close to the best-performing classical methods, demonstrating its practical suitability for reservoir decision support.

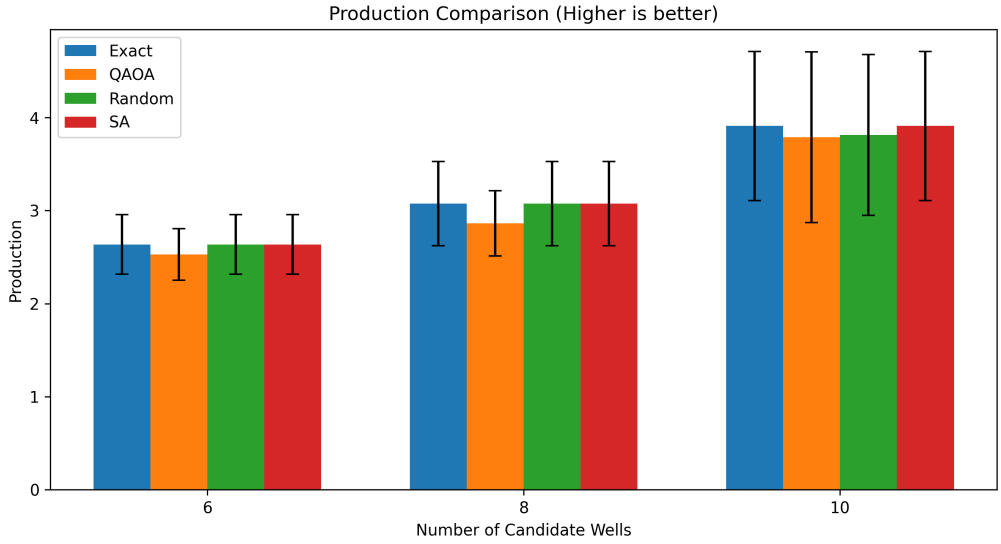

Figure 2: Production comparison across optimization methods (higher is better).

5.3 Interference Penalty Comparison

Figure 3 shows the interference penalty comparison. Lower values are preferred. QAOA demonstrates competitive interference control, which suggests that it can effectively balance spatial inter-

actions among selected wells.

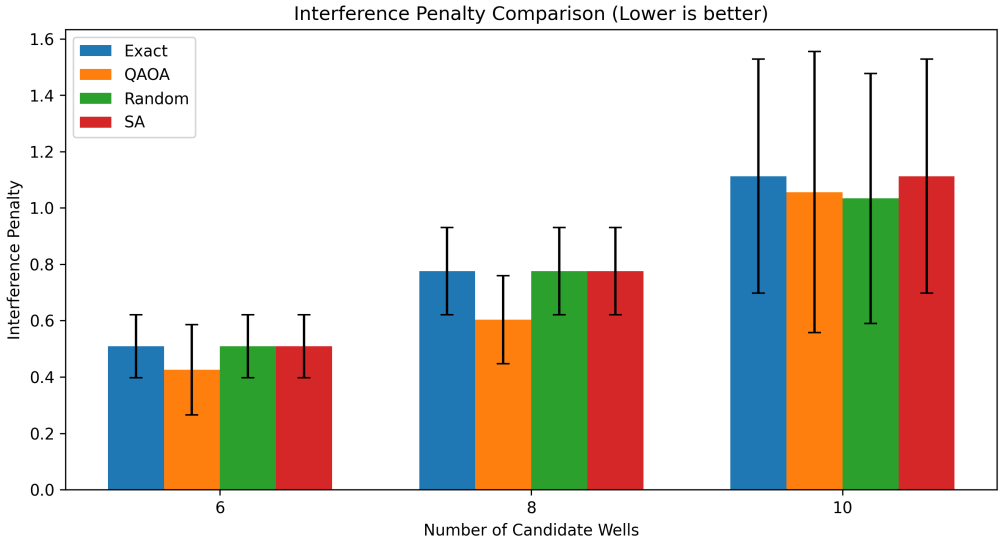

Figure 3: Interference penalty comparison across optimization methods (lower is better).

5.4 Runtime Comparison

Figure 4 compares runtime performance. For small-scale instances, exact optimization remains computationally efficient. However, its complexity grows exponentially with problem size, whereas QAOA provides a more scalable framework for future extension to larger systems.

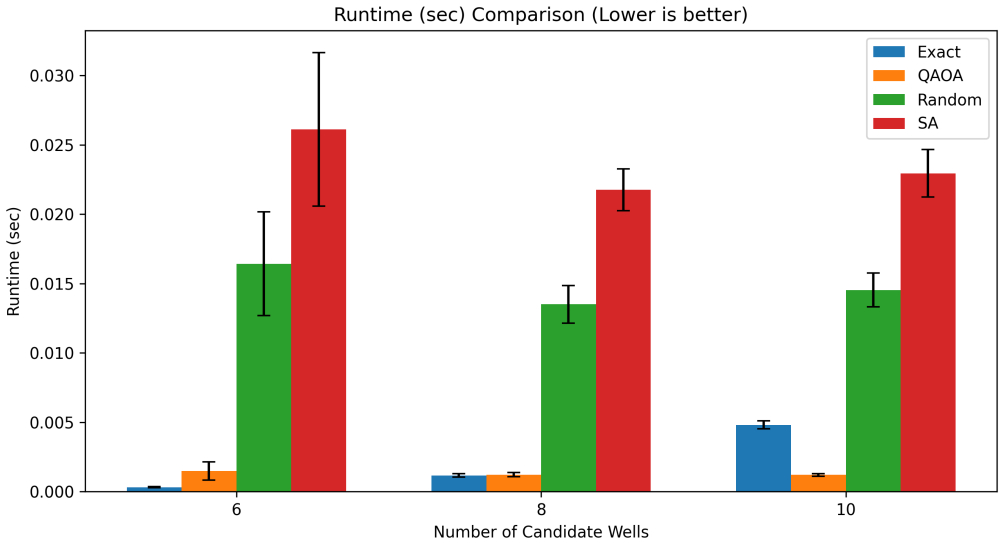

Figure 4: Runtime comparison across optimization methods (lower is better).

5.5 Numerical Summary

Table 1 summarizes representative numerical results extracted from the benchmarking experiments.

Table 1: Representative comparison of optimization methods.

Method	Number of Wells	Energy	Production	Runtime (sec)
Exact	6	-2.16	2.71	0.0003
QAOA	6	-2.16	2.71	0.0015
Random	6	-2.16	2.71	0.0160
SA	6	-2.16	2.71	0.0260
Exact	8	-2.30	3.07	0.0011
QAOA	8	-2.25	2.86	0.0012
Random	8	-2.30	3.07	0.0135
SA	8	-2.30	3.07	0.0220
Exact	10	-2.80	3.90	0.0047
QAOA	10	-2.72	3.78	0.0012
Random	10	-2.78	3.82	0.0145
SA	10	-2.80	3.90	0.0230

5.6 Discussion

The results indicate that QAOA can achieve near-optimal performance in terms of objective energy while preserving strong production and interference characteristics. For the tested small-scale instances, exact optimization remains highly competitive in runtime, which is expected given the limited search space. However, this advantage is not expected to hold as the number of candidate wells increases.

The practical value of the proposed framework lies in its scalability pathway. As reservoir optimization problems grow larger and more realistic, exhaustive search becomes infeasible, while hybrid quantum-classical strategies remain attractive due to their ability to approximate high-quality solutions without full combinatorial enumeration.

6 Conclusion

This study presented a scalable hybrid quantum-classical framework for oil reservoir well placement optimization using QUBO modeling and QAOA. The proposed approach successfully integrates production-interference trade-off modeling with quantum-inspired optimization.

The results show that QAOA achieves near-optimal solutions and competitive production behavior, while also maintaining effective interference control. Although exact methods remain faster for small problem sizes, the hybrid framework offers a promising foundation for future large-scale reservoir optimization.

Future work will focus on real reservoir datasets, stronger physical constraints, deeper QAOA circuits, and deployment on advanced quantum hardware.

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